Aurora: Module Relay x8

Specifications

Prepared for

Engineering Team

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
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Table of Contents

Revision History 2

Table of Contents 3

1. Specifications 14

1.1 CORE Communication 15

1.1.2 Hardware 15

1.1.1 Software 16

1.2 CORE Commands 17

1.2.1 CORE Command: Turn ON a realy driver 18

1.2.1.1 CORE Command: Turn ON a realy driver number 1 19

1.2.1.2 CORE Command: Turn ON a realy driver number 2 20

1.2.1.3 CORE Command: Turn ON a realy driver number 3 21

1.2.1.4 CORE Command: Turn ON a realy driver number 4 22

1.2.1.5 CORE Command: Turn ON a realy driver number 5 23

1.2.1.6 CORE Command: Turn ON a realy driver number 6 24

1.2.1.7 CORE Command: Turn ON a realy driver number 7 25

1.2.1.8 CORE Command: Turn ON a realy driver number 8 26

1.2.2 CORE Command: Turn OFF a realy driver 27

1.2.2.1 CORE Command: Turn OFF a realy driver number 1 28

1.2.2.2 CORE Command: Turn OFF a realy driver number 2 29

1.2.2.3 CORE Command: Turn OFF a realy driver number 3 30

1.2.2.4 CORE Command: Turn OFF a realy driver number 4 31

1.2.2.5 CORE Command: Turn OFF a realy driver number 5 32

1.2.2.6 CORE Command: Turn OFF a realy driver number 6 33

1.2.2.7 CORE Command: Turn OFF a realy driver number 7 34

1.2.2.8 CORE Command: Turn OFF a realy driver number 8 35

1.3 CORE Report 36
1.3.1 CORE Report: Status 37
1.3.1.1 CORE Report: Status INITIALIZING 39
1.3.1.2 CORE Report: Status READY 40
1.3.1.3 CORE Report: Status OPERATING 41
1.3.1.4 CORE Report: Status WARNING 42
1.3.1.5 CORE Report: Status FAULT 43
1.3.1.6 CORE Report: Status MAINTENANCE 44
1.3.1.7 CORE Report: Status OFFLINE 45
1.3.1 CORE Report: Status INITIALIZING 46
1.2 CORE Execute 46
1.3 CORE Report 47
1.3.1 The Relay Module shall report to CORE its status 48
Statements 48
1.3.2 The Relay Module shall report to CORE outputs status 49
Statements 49
1.3.3 The Relay Module shall report to CORE communication status 50
Statements 50
1.3.4 The Relay Module shall report to CORE diagnostics 51
Statements 51
1.3.5 The Relay Module shall report to CORE diagnostics identification 52
Statements 52
1.3.6 The Relay Module shall report to CORE fault information 53
Statements 53
1.3.7 The Relay Module shall report to CORE individual channel faults 54
Statements 54
1.3.8 The Relay Module shall report to CORE any condition preventing reliable output control . . . 55

Statements . . . 55
1.3.9 The Relay Module shall report to CORE module health . . . 56
Statements . . . 56
1.3.10 The Relay Module shall report to CORE electrical identification . . . 57
Statements . . . 57
1.3.11 The Relay Module shall report to CORE thermal identification . . . 58
Statements . . . 58
1.3.12 The Relay Module shall report to CORE firmware identification . . . 59
Statements . . . 59
1.3.13 The Relay Module shall report to CORE hardware identification . . . 60
Statements . . . 60
1.4 CORE Monitor . . . 61
1.4.1 The Relay Module shall monitor communication timeout with CORE . . . 62
Statements . . . 62
1.4.2 The Relay Module shall monitor timeout with CORE to transition to its outputs to predefined safe state . . . 63
Statements . . . 63
2. Module 64
2.1 Module Relay Driver . . . 65
2.1.1 The Relay Module shall provide eight indipendently controlled relay-driven channels . . . 66
Statements . . . 66
2.1.2 The Relay Module shall manage activation of each driver channel . . . 67
Statements . . . 67
2.1.3 The Relay Module shall manage deactivation of each driver channel . . . 68
Statements . . . 68
2.1.4 The Relay Module shall energize each relay coil independently across the full specified supply 69
Statements . . . 69

2.1.5 The Relay Module shall energize its assigned relay coil across the full specified temperature tolerance range . . . 70

Statements . . . 70

2.1.6 The Relay Module shall energize its assigned relay coil with adequate voltage margin . . . 71

Statements . . . 71

2.1.7 The Relay Module shall energize its assigned relay coil with adequate current margin . . . 72

Statements . . . 72

2.1.8 The Relay Module shall energize its assigned relay coil with continuous-duty capability . . . 73

Statements . . . 73

2.1.9 The Relay Module shall energize its assigned relay coil with short-circuit protection . . . 74

Statements . . . 74

2.1.10 The Relay Module shall energize its assigned relay coil with overload protection . . . 75

Statements . . . 75

2.1.11 The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions . . . 76

Statements . . . 76

2.1.12 The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions . . . 77

Statements . . . 77

2.1.13 The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings . . . 78

Statements . . . 78

2.1.14 The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays . . . 79

Statements . . . 79

2.1.15 The Relay Module shall have drivers with appropriately sized suppressions . . . 80

Statements . . . 80

2.1.16 The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode . . . 81

Statements . . . 81

2.1.17 The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network . . . 82

Statements . . . 82

2.2 Module Power Supply . . . 83

2.2.1 The Relay Module shall operate correctly over the complete specified nominal supply range 84

Statements . . . 84

2.2.2 The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances . . . 85

Statements . . . 85

2.2.3 The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation . . . 86

Statements . . . 86

2.2.4 The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions . . . 87

Statements . . . 87

2.2.5 The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts . . . 88

Statements . . . 88

2.2.6 The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage . . . 89

Statements . . . 89

2.2.7 The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple . . . 90

Statements . . . 90

2.2.8 The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet . . . 91

Statements . . . 91

2.2.9 The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple . . . 92

Statements . . . 92

2.3 Module Startup . . . 93

2.3.1 The Relay Module shall include a startup sequence that verifies all required permissives . . 94

Statements . . . 94

2.3.2 The Relay Module shall include a startup sequence that establishes the required gas flow . . 95

Statements . . . 95

2.3.3 The Relay Module shall include a startup sequence that establishes the supporting utilities . 96

Statements . . . 96

2.3.4 The Relay Module shall include a startup sequence that enables cooling . . . 97

Statements . . . 97

2.3.5 The Relay Module shall include a startup sequence that enables other auxiliaries . . . 98

Statements . . . 98

2.3.6 The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries . . . 99

Statements . . . 99

2.3.7 The Relay Module shall include a startup sequence that energizes the ozone-generation system

Statements . . . 100

2.3.8 The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete . . . 101

Statements . . . 101

2.3.9 The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration . . . 102

Statements . . . 102

2.4 Module Shutdown . . . 103

2.4.1 The Relay Module shall include a shutdown sequence that disables ozone generation . . . 104

Statements . . . 104

2.4.2 The Relay Module shall include a shutdown sequence that maintains the required cooling . 105

Statements . . . 105

2.4.3 The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period . . . 106

Statements . . . 106

2.4.4 The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order . . . 107

Statements . . . 107

2.4.5 The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state . . . 108

Statements . . . 108

2.5 Module Permit . . . 109

2.5.1 The Relay Module shall permit ozone generation only when all required utilities are available 110

Statements . . . 110

2.5.2 The Relay Module shall permit ozone generation only when gas flow is within their specified limits . . . 111

Statements . . . 111

2.5.3 The Relay Module shall permit ozone generation only when cooling is within their specified limits . . . 112

Statements . . . 112

2.5.4 The Relay Module shall permit ozone generation only when required valves are in their correct states . . . 113

Statements . . . 113

2.5.5 The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states . . . 114

Statements . . . 114

2.5.6 The Relay Module shall permit ozone generation only when no active safety interlock is violated . . . 115

Statements . . . 115

2.5.7 The Relay Module shall permit ozone generation only when no active process interlock is violated . . . 116

Statements . . . 116

2.5.8 The Relay Module shall permit ozone generation only when all required feedback signals are valid . . . 117

Statements . . . 117

2.5.9 The Relay Module shall permit ozone generation only when the system is in an authorized operating mode . . . 118

Statements . . . 118

2.6 Module Stop . . . 119

2.6.1 The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected . . . 120

Statements . . . 120

2.6.2 The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow . . . 121

Statements . . . 121

2.6.3 The Relay Module shall immediately stop ozone generation when detected a loss of required cooling . . . 122

Statements . . . 122

2.6.4 The Relay Module shall immediately stop ozone generation when detected a critical equipment failure . . . 123

Statements . . . 123

2.6.5 The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits . . . 124

Statements . . . 124

2.6.6 The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives . . . 125

Statements . . . 125

2.6.7 The Relay Module shall immediately stop ozone generation when detected emergency shutdown . . . 126

Statements . . . 126

2.6.8 The Relay Module shall immediately stop ozone generation when detected defined critical interlock . . . 127

Statements . . . 127

2.7 Module Enforce . . . 128

2.7.1 The Relay Module shall locally enforce fault-handling . . . 129

Statements . . . 129

2.7.2 The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels . . . 130

Statements . . . 130

2.7.3 The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels . . 131

Statements . . . 131

2.7.4 The Relay Module shall locally enforce output-level safety constraints . . . 132

Statements . . . 132

2.7.5 The Relay Module shall locally enforce watchdog/communication-loss responses . . . 133

Statements . . . 133

2.8 Module Diagnostic . . . 134

2.8.1 The Relay Module shall locally perform hardware diagnostics . . . 135

Statements . . . 135

2.9 Module Safety . . . 136

2.9.1 The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout . . . 137

Statements . . . 137

2.9.2 The Relay Module shall transition all relay outputs to defined safe state during power loss . 138

Statements . . . 138

2.9.3 The Relay Module shall define safe-states for all its outputs (based on its associated hazard analysis) . . . 139

Statements . . . 139

2.9.4 The Relay Module shall define OFF/de-energized as the default safe-state unless a documented safety analysis requires a specific output to remain energized . . . 140

Statements . . . 140

2.9.5 The Relay Module shall transition its outputs to safe states when system-level fault . . . 141

Statements . . . 141

2.10 Module Fault . . . 142

2.10.1 The Relay Module shall monitor output faults . . . 143

Statements . . . 143

2.10.2 The Relay Module shall maintain the output defined safe behavior under fault conditions . 144

Statements . . . 144

2.10.3 The Relay Module shall prevent one faulty channel to affect other channels 145

Statements 145

2.10.4 The Relay Module shall detect relevant hardware faults 146

Statements 146

2.10.5 The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation 147

Statements 147

3. Relay 148

3.1 Relay Coil 149

3.1.1 The Relay Module shall operate with coils that have a defined nominal voltage range . . . 150

Statements 150

3.1.2 The Relay Module shall operate with coils that have defined nominal current 151

Statements 151

3.1.3 The Relay Module shall operate with coils that have defined maximum current 152

Statements 152

3.1.4 The Relay Module shall operate with coils that have defined resistance 153

Statements 153

3.1.5 The Relay Module shall operate with coils that have defined inductance 154

Statements 154

3.1.6 The Relay Module shall operate with coils that have defined inrush behavior 155

Statements 155

3.1.7 The Relay Module shall operate with coils that have defined continuous-duty rating 156

Statements 156

3.1.8 The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver 157

Statements 157

3.1.9 The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver 158

Statements 158

1. Specifications

1.1 CORE Communication

1.1.2 Hardware

The Relay Module shall communicate with CORE via RS-485 physical interface.

Setting	Value
Physical interface	RS-485
Number of wires	2-wire (A, B)

1.1.1 Software

The Relay Module shall communicate with CORE via Modbus RTU protocol.

Setting	Value
Protocollo	Modbus RTU
Module role	Slave
Module ID	0x01
Baud rate	9600 bps
Data format	8 bits, no parity, 1 stop bit (8N1)

1.2 CORE Commands

The Relay Module shall execute commands received from CORE to turn ON/OFF each relay driver independently.

1.2.1 CORE Command: Turn ON a realy driver

The Relay Module shall turn ON a relay driver when receiving one of the following Modbus commands.

1.2.1.1 CORE Command: Turn ON a realy driver number 1

The Relay Module shall turn ON the relay driver number 1 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x00
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 00 FF 00 XX XX

1.2.1.2 CORE Command: Turn ON a realy driver number 2

The Relay Module shall turn ON the relay driver number 2 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x01
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 01 FF 00 XX XX

1.2.1.3 CORE Command: Turn ON a realy driver number 3

The Relay Module shall turn ON the relay driver number 3 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x02
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 02 FF 00 XX XX

1.2.1.4 CORE Command: Turn ON a realy driver number 4

The Relay Module shall turn ON the relay driver number 4 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x03
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 03 FF 00 XX XX

1.2.1.5 CORE Command: Turn ON a realy driver number 5

The Relay Module shall turn ON the relay driver number 5 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x04
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 04 FF 00 XX XX

1.2.1.6 CORE Command: Turn ON a realy driver number 6

The Relay Module shall turn ON the relay driver number 6 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x05
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 05 FF 00 XX XX

1.2.1.7 CORE Command: Turn ON a realy driver number 7

The Relay Module shall turn ON the relay driver number 7 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x06
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 06 FF 00 XX XX

1.2.1.8 CORE Command: Turn ON a realy driver number 8

The Relay Module shall turn ON the relay driver number 8 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x07
Turn ON	0xFF 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 07 FF 00 XX XX

1.2.2 CORE Command: Turn OFF a realy driver

The Relay Module shall turn OFF a relay driver when receiving one of the following Modbus commands.

1.2.2.1 CORE Command: Turn OFF a realy driver number 1

The Relay Module shall turn OFF the relay driver number 1 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x00
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 00 FF 00 XX XX

1.2.2.2 CORE Command: Turn OFF a realy driver number 2

The Relay Module shall turn OFF the relay driver number 2 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x01
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 01 FF 00 XX XX

1.2.2.3 CORE Command: Turn OFF a realy driver number 3

The Relay Module shall turn OFF the relay driver number 3 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x02
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 02 FF 00 XX XX

1.2.2.4 CORE Command: Turn OFF a realy driver number 4

The Relay Module shall turn OFF the relay driver number 4 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x03
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 03 FF 00 XX XX

1.2.2.5 CORE Command: Turn OFF a realy driver number 5

The Relay Module shall turn OFF the relay driver number 5 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x04
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 04 FF 00 XX XX

1.2.2.6 CORE Command: Turn OFF a realy driver number 6

The Relay Module shall turn OFF the relay driver number 6 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x05
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 05 FF 00 XX XX

1.2.2.7 CORE Command: Turn OFF a realy driver number 7

The Relay Module shall turn OFF the relay driver number 7 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x06
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 06 FF 00 XX XX

1.2.2.8 CORE Command: Turn OFF a realy driver number 8

The Relay Module shall turn OFF the relay driver number 8 when receiving the following Modbus commands.

Description	Value
Module ID	0x01
Function code	0x05
Coil address	0x00 0x07
Turn OFF	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 05 00 07 FF 00 XX XX

1.3 CORE Report

1.3.1 CORE Report: Status

The Relay Module shall report to CORE its status.

Modbus register definition:

Parameter	Value
Register name	Module_Status
Register address	[TODO] 30001 (0x0000 zero-based)
Function code	FC04
Register type	Input Register
Data type	UINT16
Access	Read-only
Default value	0 (INITIALIZING)
Update	Whenever operational state changes

Modbus master command:

Parameter	Value
Module ID	0x01
Function code	0x04
Starting address	[TODO] 0x00 0x00
Quantity	[TODO] 0x00 0x01
CRC16	0xXX 0xXX

The complete data frame is: 01 04 00 00 00 01 XX XX

Modbus slave response:

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02

Parameter	Value
Register value	0xXX 0xXX
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 XX XX XX XX

1.3.1.1 CORE Report: Status INITIALIZING

The Relay Module shall report the status INITIALIZING (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x00
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 00 XX XX

1.3.1.2 CORE Report: Status READY

The Relay Module shall report the status READY (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x01
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 01 XX XX

1.3.1.3 CORE Report: Status OPERATING

The Relay Module shall report the status OPERATING (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x02
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 02 XX XX

1.3.1.4 CORE Report: Status WARNING

The Relay Module shall report the status WARNING (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x03
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 03 XX XX

1.3.1.5 CORE Report: Status FAULT

The Relay Module shall report the status FAULT (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x04
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 04 XX XX

1.3.1.6 CORE Report: Status MAINTENANCE

The Relay Module shall report the status MAINTENANCE (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x05
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 05 XX XX

1.3.1.7 CORE Report: Status OFFLINE

The Relay Module shall report the status OFFLINE (if it's in such state) when Modbus asks for it.

Parameter	Value
Module ID	0x01
Function code	0x04
Number of data bytes	0x00 0x02
Register value	0x00 0x06
CRC16	0xXX 0xXX

The complete data frame is: 01 04 02 00 06 XX XX

Modbus register definition

Parameter Definition

Register name Module_Status

Register address 30001 (example)

Function code FC04

Register type Input Register

Data type UINT16

Access Read-only

Default value 0 (INITIALIZING)

Update Whenever operational state changes

The status is a UINT16 Modbus register containing an enumerated state.

Module_Status - UINT16 - Read-only

1.3.1 CORE Report: Status INITIALIZING

Value State Meaning

0 INITIALIZING Module is powering up and performing initialization.

1 READY Module is initialized and ready to operate.

2 OPERATING Module is operating and controlling its outputs.

3 WARNING Module is operational but has an active warning.

4 FAULT A fault has been detected that affects normal operation.

5 MAINTENANCE Module is in maintenance or service mode.

6 OFFLINE Module is not operational or has been intentionally disabled.

1.2 CORE Execute

- The Relay Module shall execute relay-output commands (ON/OFF) deterministically
- The Relay Module shall execute relay-output commands (ON/OFF) within specified time
- The Relay Module shall execute normal operation only when explicit reauthorization from CORE is given

1.3 CORE Report

- The Relay Module shall report to CORE its status
- The Relay Module shall report to CORE outputs status
- The Relay Module shall report to CORE communication status
- The Relay Module shall report to CORE diagnostics
- The Relay Module shall report to CORE diagnostics identification
- The Relay Module shall report to CORE fault information
- The Relay Module shall report to CORE individual channel faults
- The Relay Module shall report to CORE any condition preventing reliable output control
- The Relay Module shall report to CORE module health
- The Relay Module shall report to CORE electrical identification
- The Relay Module shall report to CORE thermal identification
- The Relay Module shall report to CORE firmware identification
- The Relay Module shall report to CORE hardware identification

1.3.1 The Relay Module shall report to CORE its status

Statements

- The Relay Module shall report its status to CORE

1.3.2 The Relay Module shall report to CORE outputs status

Statements

- The Relay Module shall provide CORE with output status
- The Relay Module shall communicate channels status to CORE
- The Relay Module shall report the state of all eight outputs

1.3.3 The Relay Module shall report to CORE communication status

Statements

- The Relay Module shall report communication status
- The Relay Module shall report the communication fault when communication with CORE is restored

1.3.4 The Relay Module shall report to CORE diagnostics

Statements

- The Relay Module shall provide CORE with module diagnostics
- The Relay Module shall communicate channels diagnostics to CORE

1.3.5 The Relay Module shall report to CORE diagnostics identification

Statements

- The Relay Module shall report diagnostics identification

1.3.6 The Relay Module shall report to CORE fault information

Statements

- The Relay Module shall provide sufficient fault information to CORE when in failure state
- The Relay Module shall report its faults to CORE
- The Relay Module shall report the fault as required when it detects an internal fault capable of compromising safe operation

1.3.7 The Relay Module shall report to CORE individual channel faults

Statements

- The Relay Module shall report individual channel faults

1.3.8 The Relay Module shall report to CORE any condition preventing reliable output control

Statements

- The Relay Module shall report any condition preventing reliable output control

1.3.9 The Relay Module shall report to CORE module health

Statements

- The Relay Module shall report the module health

1.3.10 The Relay Module shall report to CORE electrical identification

Statements

- The Relay Module shall report electrical identification

1.3.11 The Relay Module shall report to CORE thermal identification

Statements

- The Relay Module shall report thermal identification

1.3.12 The Relay Module shall report to CORE firmware identification

Statements

- The Relay Module shall report firmware identification

1.3.13 The Relay Module shall report to CORE hardware identification

Statements

- The Relay Module shall report hardware identification

1.4 CORE Monitor

- The Relay Module shall monitor communication timeout with CORE
- The Relay Module shall monitor timeout with CORE to transition to its outputs to predefined safe state

1.4.1 The Relay Module shall monitor communication timeout with CORE

Statements

- The Relay Module shall detect communication timeout with CORE

1.4.2 The Relay Module shall monitor timeout with CORE to transition to its outputs to predefined safe state

Statements

- The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE
- The Relay Module shall autonomously transition every output to its predefined safe state when communication with the CORE is lost beyond the defined safety timeout
- The Relay Module shall maintain a predefined safe state when communication with the CORE is lost beyond the defined safety timeout

2. Module

2.1 Module Relay Driver

- The Relay Module shall provide eight indipendently controlled relay-driven channels
- The Relay Module shall manage activation of each driver channel
- The Relay Module shall manage deactivation of each driver channel
- The Relay Module shall energize each relay coil independently across the full specified supply
- The Relay Module shall energize its assigned relay coil across the full specified temperature tolerance range
- The Relay Module shall energize its assigned relay coil with adequate voltage margin
- The Relay Module shall energize its assigned relay coil with adequate current margin
- The Relay Module shall energize its assigned relay coil with continuous-duty capability
- The Relay Module shall energize its assigned relay coil with short-circuit protection
- The Relay Module shall energize its assigned relay coil with overload protection
- The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions
- The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions
- The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings
- The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays
- The Relay Module shall have drivers with appropriately sized suppressions
- The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode
- The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

2.1.1 The Relay Module shall provide eight indipendently controlled relay-driven channels

Statements

- The Relay Module shall provide eight indipendently controlled relay-driven channels
- The Relay Module shall manage its eight physical driver outputs

2.1.2 The Relay Module shall manage activation of each driver channel

Statements

- The Relay Module shall manage activation of each channel

2.1.3 The Relay Module shall manage deactivation of each driver channel

Statements

- The Relay Module shall manage deactivation of each channel

2.1.4 The Relay Module shall energize each relay coil independently across the full specified supply

Statements

- The Relay Module shall reliably energize its assigned relay coil across the full specified supply
- The Relay Module shall independently energize its assigned relay coil across the full specified supply

2.1.5 The Relay Module shall energize its assigned relay coil across the full specified temperature tolerance range

Statements

- The Relay Module shall energize its assigned relay coil across the full specified temperature
- The Relay Module shall energize its assigned relay coil across the full specified tolerance range

2.1.6 The Relay Module shall energize its assigned relay coil with adequate voltage margin

Statements

- The Relay Module shall energize its assigned relay coil with adequate voltage
- The Relay Module shall energize its assigned relay coil with adequate voltage margin

2.1.7 The Relay Module shall energize its assigned relay coil with adequate current margin

Statements

- The Relay Module shall energize its assigned relay coil with adequate current margin

2.1.8 The Relay Module shall energize its assigned relay coil with continuous-duty capability

Statements

- The Relay Module shall energize its assigned relay coil with continuous-duty
- The Relay Module shall energize its assigned relay coil with continuous-duty capability

2.1.9 The Relay Module shall energize its assigned relay coil with short-circuit protection

Statements

- The Relay Module shall energize its assigned relay coil with short-circuit protection

2.1.10 The Relay Module shall energize its assigned relay coil with overload protection

Statements

- The Relay Module shall energize its assigned relay coil with overload protection

2.1.11 The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

Statements

- The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

2.1.12 The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

Statements

- The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

2.1.13 The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

Statements

- The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

2.1.14 The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

Statements

- The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

2.1.15 The Relay Module shall have drivers with appropriately sized suppressions

Statements

- The Relay Module shall have drivers with appropriately sized suppressions

2.1.16 The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

Statements

- The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

2.1.17 The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Statements

- The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

2.2 Module Power Supply

- The Relay Module shall operate correctly over the complete specified nominal supply range
- The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances
- The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation
- The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions
- The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts
- The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage
- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple
- The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet
- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

2.2.1 The Relay Module shall operate correctly over the complete specified nominal supply range

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range

2.2.2 The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

2.2.3 The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

2.2.4 The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

2.2.5 The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

2.2.6 The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

2.2.7 The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

2.2.8 The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

2.2.9 The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

2.3 Module Startup

- The Relay Module shall include a startup sequence that verifies all required permissives
- The Relay Module shall include a startup sequence that establishes the required gas flow
- The Relay Module shall include a startup sequence that establishes the supporting utilities
- The Relay Module shall include a startup sequence that enables cooling
- The Relay Module shall include a startup sequence that enables other auxiliaries
- The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries
- The Relay Module shall include a startup sequence that energizes the ozone-generation system
- The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete
- The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

2.3.1 The Relay Module shall include a startup sequence that verifies all required permissives

Statements

- The Relay Module shall include a startup sequence that verifies all required permissives

2.3.2 The Relay Module shall include a startup sequence that establishes the required gas flow

Statements

- The Relay Module shall include a startup sequence that establishes the required gas flow

2.3.3 The Relay Module shall include a startup sequence that establishes the supporting utilities

Statements

- The Relay Module shall include a startup sequence that establishes the supporting utilities

2.3.4 The Relay Module shall include a startup sequence that enables cooling

Statements

- The Relay Module shall include a startup sequence that enables cooling

2.3.5 The Relay Module shall include a startup sequence that enables other auxiliaries

Statements

- The Relay Module shall include a startup sequence that enables other auxiliaries

2.3.6 The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

Statements

- The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

2.3.7 The Relay Module shall include a startup sequence that energizes the ozone-generation system

Statements

- The Relay Module shall include a startup sequence that energizes the ozone-generation system

2.3.8 The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

Statements

- The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

2.3.9 The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

Statements

- The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

2.4 Module Shutdown

- The Relay Module shall include a shutdown sequence that disables ozone generation
- The Relay Module shall include a shutdown sequence that maintains the required cooling
- The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period
- The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order
- The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

2.4.1 The Relay Module shall include a shutdown sequence that disables ozone generation

Statements

- The Relay Module shall include a shutdown sequence that disables ozone generation

2.4.2 The Relay Module shall include a shutdown sequence that maintains the required cooling

Statements

- The Relay Module shall include a shutdown sequence that maintains the required cooling

2.4.3 The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

Statements

- The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

2.4.4 The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

Statements

- The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

2.4.5 The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

Statements

- The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

2.5 Module Permit

- The Relay Module shall permit ozone generation only when all required utilities are available
- The Relay Module shall permit ozone generation only when gas flow is within their specified limits
- The Relay Module shall permit ozone generation only when cooling is within their specified limits
- The Relay Module shall permit ozone generation only when required valves are in their correct states
- The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states
- The Relay Module shall permit ozone generation only when no active safety interlock is violated
- The Relay Module shall permit ozone generation only when no active process interlock is violated
- The Relay Module shall permit ozone generation only when all required feedback signals are valid
- The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

2.5.1 The Relay Module shall permit ozone generation only when all required utilities are available

Statements

- The Relay Module shall permit ozone generation only when all required utilities are available

2.5.2 The Relay Module shall permit ozone generation only when gas flow is within their specified limits

Statements

- The Relay Module shall permit ozone generation only when gas flow is within their specified limits

2.5.3 The Relay Module shall permit ozone generation only when cooling is within their specified limits

Statements

- The Relay Module shall permit ozone generation only when cooling is within their specified limits

2.5.4 The Relay Module shall permit ozone generation only when required valves are in their correct states

Statements

- The Relay Module shall permit ozone generation only when required valves are in their correct states

2.5.5 The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states

Statements

- The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states

2.5.6 The Relay Module shall permit ozone generation only when no active safety interlock is violated

Statements

- The Relay Module shall permit ozone generation only when no active safety interlock is violated

2.5.7 The Relay Module shall permit ozone generation only when no active process interlock is violated

Statements

- The Relay Module shall permit ozone generation only when no active process interlock is violated

2.5.8 The Relay Module shall permit ozone generation only when all required feedback signals are valid

Statements

- The Relay Module shall permit ozone generation only when all required feedback signals are valid

2.5.9 The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

Statements

- The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

2.6 Module Stop

- The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected
- The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow
- The Relay Module shall immediately stop ozone generation when detected a loss of required cooling
- The Relay Module shall immediately stop ozone generation when detected a critical equipment failure
- The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits
- The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives
- The Relay Module shall immediately stop ozone generation when detected emergency shutdown
- The Relay Module shall immediately stop ozone generation when detected defined critical interlock

2.6.1 The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

Statements

- The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

2.6.2 The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

Statements

- The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

2.6.3 The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

Statements

- The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

2.6.4 The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

Statements

- The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

2.6.5 The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

Statements

- The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

2.6.6 The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

Statements

- The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

2.6.7 The Relay Module shall immediately stop ozone generation when detected emergency shutdown

Statements

- The Relay Module shall immediately stop ozone generation when detected emergency shutdown

2.6.8 The Relay Module shall immediately stop ozone generation when detected defined critical interlock

Statements

- The Relay Module shall immediately stop ozone generation when detected defined critical interlock

2.7 Module Enforce

- The Relay Module shall locally enforce fault-handling
- The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels
- The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels
- The Relay Module shall locally enforce output-level safety constraints
- The Relay Module shall locally enforce watchdog/communication-loss responses

2.7.1 The Relay Module shall locally enforce fault-handling

Statements

- The Relay Module shall enforce local fault-handling
- The Relay Module shall locally enforce internal fault handling

2.7.2 The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

Statements

- The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

2.7.3 The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

Statements

- The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

2.7.4 The Relay Module shall locally enforce output-level safety constraints

Statements

- The Relay Module shall locally enforce output-level safety constraints

2.7.5 The Relay Module shall locally enforce watchdog/communication-loss responses

Statements

- The Relay Module shall locally enforce watchdog/communication-loss responses

2.8 Module Diagnostic

- The Relay Module shall locally perform hardware diagnostics

2.8.1 The Relay Module shall locally perform hardware diagnostics

Statements

- The Relay Module shall perform local hardware diagnostics

2.9 Module Safety

- The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout
- The Relay Module shall transition all relay outputs to defined safe state during power loss
- The Relay Module shall define safe-states for all its outputs (based on its associated hazard analysis)
- The Relay Module shall define OFF/de-energized as the default safe-state unless a documented safety analysis requires a specific output to remain energized
- The Relay Module shall transition its outputs to safe states when system-level fault

2.9.1 The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Statements

- The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

2.9.2 The Relay Module shall transition all relay outputs to defined safe state during power loss

Statements

- The Relay Module shall transition all relay outputs to defined safe state during power loss

2.9.3 The Relay Module shall define safe-states for all its outputs (based on its associated hazard analysis)

Statements

- The Relay Module shall have a formally defined safe state for all its outputs based on its associated hazard analysis

2.9.4 The Relay Module shall define OFF/de-energized as the default safe-state unless a documented safety analysis requires a specific output to remain energized

Statements

- The Relay Module shall have OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized

2.9.5 The Relay Module shall transition its outputs to safe states when system-level fault

Statements

- The Relay Module shall transition its outputs to safe states when system-level fault
- The Relay Module shall transition to defined safe state when in failure state

2.10 Module Fault

- The Relay Module shall monitor output faults
- The Relay Module shall maintain the output defined safe behavior under fault conditions
- The Relay Module shall prevent one faulty channel to affect other channels
- The Relay Module shall detect relevant hardware faults
- The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

2.10.1 The Relay Module shall monitor output faults

Statements

- The Relay Module shall monitor output faults
- The Relay Module shall detect relevant output faults

2.10.2 The Relay Module shall maintain the output defined safe behavior under fault conditions

Statements

- The Relay Module shall maintain the output defined safe behavior under fault conditions
- The Relay Module shall prevent unintended activation when in failure state
- The Relay Module shall transition affected outputs to their predefined safe states independently of higher-level system decisions
- The Relay Module shall immediately prevent unsafe output activation when it detects an internal fault capable of compromising safe operation
- The Relay Module shall force all affected outputs to their defined safe states when it detects an internal fault capable of compromising safe operation
- The Relay Module shall latch the fault as required when it detects an internal fault capable of compromising safe operation
- The Relay Module shall prevent reactivation until the fault-clear conditions are satisfied when it detects an internal fault capable of compromising safe operation
- The Relay Module shall locally enforce defined safe-state behavior without relying solely on the CORE

2.10.3 The Relay Module shall prevent one faulty channel to affect other channels

Statements

- The Relay Module shall prevent one faulty channel to affect other channels

2.10.4 The Relay Module shall detect relevant hardware faults

Statements

- The Relay Module shall detect relevant hardware faults

2.10.5 The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

Statements

- The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

3. Relay

3.1 Relay Coil

- The Relay Module shall operate with coils that have a defined nominal voltage range
- The Relay Module shall operate with coils that have defined nominal current
- The Relay Module shall operate with coils that have defined maximum current
- The Relay Module shall operate with coils that have defined resistance
- The Relay Module shall operate with coils that have defined inductance
- The Relay Module shall operate with coils that have defined inrush behavior
- The Relay Module shall operate with coils that have defined continuous-duty rating
- The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver
- The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

3.1.1 The Relay Module shall operate with coils that have a defined nominal voltage range

Statements

- The Relay Module shall operate with coils that have a defined nominal voltage
- The Relay Module shall operate with coils that have allowable voltage range

3.1.2 The Relay Module shall operate with coils that have defined nominal current

Statements

- The Relay Module shall operate with coils that have nominal current

3.1.3 The Relay Module shall operate with coils that have defined maximum current

Statements

- The Relay Module shall operate with coils that have maximum current

3.1.4 The Relay Module shall operate with coils that have defined resistance

Statements

- The Relay Module shall operate with coils that have defined resistance

3.1.5 The Relay Module shall operate with coils that have defined inductance

Statements

- The Relay Module shall operate with coils that have defined inductance

3.1.6 The Relay Module shall operate with coils that have defined inrush behavior

Statements

- The Relay Module shall operate with coils that have defined inrush behavior

3.1.7 The Relay Module shall operate with coils that have defined continuous-duty rating

Statements

- The Relay Module shall operate with coils that have defined continuous-duty rating

3.1.8 The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Statements

- The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

3.1.9 The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Statements

- The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver